

Week 3 — Building the Analytic Cohort: Case Definitions, Eligibility, and the Funnel

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

Week 3 — Building the Analytic Cohort

Case definitions, eligibility, and the funnel

- ▶ The step that decides *who is even in the study*
- ▶ From a clean data frame to an **analytic cohort**
- ▶ Turning a prose methods paragraph into **executable logic**

MASLD: every number in the Kueh paper rides on getting *one* cohort right.

Learning objectives

- ▶ Distinguish **target**, **source**, and **analytic** populations — and locate each in a published study.
- ▶ Reconstruct the MASLD case definition: **steatosis (FLI ≥ 60)** + **1 cardiometabolic criterion**.
- ▶ Convert a prose eligibility paragraph into a reproducible **inclusion/exclusion algorithm**.
- ▶ Build and defend an **eligibility funnel** with an N at every step.
- ▶ Explain why you **never tune the cohort to hit the paper's N**.

Continuity: where this fits

- ▶ **Last week (Wk 2 → L1):** merged NHANES cycles, harmonized names, fixed messy joins → a clean person-level frame.
- ▶ **This week:** convert that frame into the **analytic cohort** the MASLD paper analyzed.
- ▶ **M0 closes today** — your memo is due at 12 PM, and your paper locks at that deadline.
- ▶ **Sets up: L2** (build the analytic dataset) and **P1** (your own eligibility funnel).

MASLD: L1 gave us clean data; today we decide who counts as a subject.

Slide 1 — Where we are in the project (M0 closing)

- ▶ Term arc: **Replicate** → **Interrogate** → **Improve** → **Defend**
— we are still in *Replicate*.
- ▶ L1 gave a clean merged NHANES frame; today we decide *who counts as a subject*.
- ▶ **M0 closing:** memos due 12 PM today; papers are final at that deadline.
- ▶ Get the cohort wrong and *every* downstream number is wrong.

MASLD: the Kueh story rides on 4 phenotype groups drawn from *one* cohort.

Slide 2 — Today's roadmap and the one big idea

- ▶ Path: **case definition** → **eligibility** → **exclusions** → **funnel** → **feasibility**.
- ▶ Big idea: **eligibility decisions are load-bearing** — analysis choices, not chores.
- ▶ Framing question: *“If two teams read the same methods paragraph, would they build the same cohort?”*

Slide 3 — Target vs source vs analytic population

- ▶ **Target** = adults with MASLD — who we want to speak about.
- ▶ **Source** = NHANES 2007–2018 respondents.
- ▶ **Analytic** = those who survive all filters *and* have measurable exposure + outcome.
- ▶ Each narrowing step trades **generalizability** for **measurability**.
- ▶ Every drop is a potential **selection bias**, not just a smaller N.
- ▶ MASLD hook: NHANES samples the US civilian population, and the analytic cohort is the MASLD subset of the **designed fasting subsample**. Foreshadow the typed funnel: every step down has a *kind*, and only some kinds are missing data.

Slide 4 — The MASLD case definition, unpacked

- ▶ MASLD (formerly **NAFLD**) = **hepatic steatosis + metabolic dysfunction**, other causes not dominant.
- ▶ Two ingredients to operationalize: **(1) steatosis, (2) 1 cardiometabolic criterion**.
- ▶ No biopsy, no imaging in NHANES → steatosis must be ascertained **non-invasively**.

MASLD: the paper uses the **Fatty Liver Index (FLI)** as its steatosis proxy — a modeling choice we scrutinize all term.

Slide 5 — Ascertaining steatosis: the Fatty Liver Index

- ▶ FLI = a published **logistic score** predicting fatty liver from routine measures.
- ▶ Inputs: **triglycerides, BMI, GGT, waist circumference** — only **triglycerides** needs a fasting lab, and that one is enough to bind the whole index.
- ▶ Output: a **0–100 score**; the paper applies the conventional **FLI 60** rule-in cut.

MASLD: FLI already contains **BMI and waist** — the same anthropometry used later to define the phenotype groups. Flag it now; it returns as the “**circularity**” threat at M1.

Slide 6 — The FLI formula, worked

- ▶ Shape: a logistic function of $0.953 \cdot \ln(\text{TG}) + 0.139 \cdot \text{BMI} + 0.718 \cdot \ln(\text{GGT}) + 0.053 \cdot \text{waist} - 15.745$, mapped to **0–100**.
- ▶ Walk one hypothetical subject through it live — note **TG and GGT enter on the log scale**.
- ▶ A **cut** on a continuous score creates a hard boundary: small measurement error near 60 **flips case status**.

MASLD: FLI 60 is both the **disease gate** and — because it needs fasting TG — the **reason the cohort shrinks** to the fasting subsample.

Slide 7 — The cardiometabolic criteria (the “M” in MASLD)

- ▶ Require **1** of: elevated **BMI/waist**, **hyperglycemia/diabetes**, **hypertension**, **dyslipidemia** (low HDL / high TG or treatment).
- ▶ Each is itself an operationalization: **which threshold? treated vs untreated? which NHANES variable?**
- ▶ Document every criterion's variable and cut in a **codebook before coding**.

MASLD: in this cohort 1 criterion is almost always met, so it rarely moves N — but you must still report it, because *your* paper's population may differ.

Slide 8 — Eligibility and exclusions are design, not chores

- ▶ Typical filters: **adults** (age 18/20), **non-pregnant**, **valid mortality linkage**, **non-missing key measures**, disease-specific exclusions.
- ▶ Each filter answers a scientific question — “*who is this study about?*” — so each needs a stated **rationale**.
- ▶ Rule: write exclusions as **executable logic** with a comment citing the paper’s sentence.

MASLD: the paper’s exclusions include competing liver-disease causes and — critically — **heavy alcohol use**, our worked example next.

Slide 9 — Worked example: the alcohol ambiguity (one sentence, three cohorts)

- ▶ The paper states its alcohol exclusion in a **single sentence** readable three ways: *which* drinks question, *which* threshold, *missing* = *drop or keep*.
- ▶ Coded three defensible ways, the same sentence yields:
- ▶ **N 4,308** — strictest reading

MASLD: this is why the reported N never quite pins down — publish your *own* coding decision explicitly rather than guess the authors' intent.

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 - ▶ A swing of **~2,000 people** from *one sentence*.

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Slide 10 — BREAK (10 min)

Restart prompt: *“Which of the three alcohol codings would you defend, and why?”*

Slide 11 — The eligibility funnel: report N at every step

- ▶ Build top-to-bottom: **source N** → **adults** → **MASLD (FLI 60 + criterion)** → **alcohol/other exclusions** → **mortality linkage** → **analytic N**.
- ▶ Report the count **removed** *and* **remaining** at *every* line.
- ▶ The deliverable is a **CONSORT-style diagram** — a reproducibility contract a reader can re-derive.

MASLD: the paper reports **N 6,300**, groups **386 / 4,541 / 769 / 604, 585 deaths**; our **full analytic file** is **6,371**, groups **391 / 4,585 / 787 / 608, 586 deaths** — and the **locked domain** is **6,048**. The gap is the finding, and none of it is interpretable without the funnel that produced it.

Slide 12 — Why you never tune the cohort to hit the paper's N

- ▶ Temptation: quietly flip a threshold until your N matches the abstract.
- ▶ That is **fitting the process to the answer** — a *garden of forking paths*.
- ▶ Correct stance: pick the **most defensible** coding a priori, report your N, and *explain* any gap.
- ▶ A **matched N with hidden tuning** is worse than an **honest mismatch you can account for**.

MASLD: the alcohol sentence alone spans ~2,000 people — “matching” the paper's N proves nothing; it may just mean you guessed their ambiguity.

Slide 13 — Peer-review principles for eligibility

- ▶ Reviewer checklist:
 - ▶ Is the **case definition** operational?
 - ▶ Is **every exclusion** executable?
 - ▶ Does the funnel's **arithmetic close**?
 - ▶ Is the **analytic N** stated and reproducible?
- ▶ Distinguish an *honest, documented* choice from an *undocumented, load-bearing* one.
- ▶ **Charity + rigor:** assume competence, but demand the numbers reconcile.

MASLD: apply the checklist to the alcohol sentence *and* the **misabeled BMI header** (a table reads “< 25” where the cut is actually 30) — small text discrepancies, real consequences for a reader rebuilding the cohort.

Slide 14 — Feasibility: can you even measure the exposure?

- ▶ FLI needs **fasting labs** → only the **fasting subsample (WTSAF)** qualifies.
- ▶ Within the **fasting subsample**, FLI is computable for **94.7%** of adults — the 5.3% who lack it are genuine item nonresponse.
- ▶ The far larger reduction, from all adults to the fasting subsample, is **designed subsampling** — WTSAF2YR exists for it.
- ▶ Feasibility check *before* committing: is the exposure computable for enough of your target population?

MASLD: make the funnel **typed** — DESIGN / ELIGIBILITY / TARGET POPULATION / OUTCOME ASCERTAINMENT / MISSINGNESS — and note that only the last one is a missing-data problem.

Slide 15 — From paper to protocol (recap, M0 closing)

- ▶ We converted a prose methods section into an explicit cohort algorithm: **definition** → **exclusions** → **funnel** → **analytic N**.
- ▶ Every decision is **documented, defensible, and re-runnable** — the standard your P1 must meet.
- ▶ Once **M0 closes** at 12 PM today, your paper's eligibility paragraph is *your* specification to decode.

Slide 16 — Bridge to today's lab (L2 / P1)

- ▶ **In L2 (Thu):** using **EpiMethods** (ehsanx.github.io/EpiMethods), build the MASLD analytic dataset from the L1 frame — code the **FLI**, apply **FLI 60 + 1 cardiometabolic criterion**, implement **one defensible alcohol coding**, and output a **funnel with N at every step**. **Do not aim it at a number:** the paper reports ~**6,300**, our full analytic file lands on **6,371**, and yours is wherever your documented steps take you.
- ▶ **P1 (your increment):** repeat the funnel for *your own* locked paper — write each inclusion/exclusion as **executable logic**, report **N at each step**, and **flag one ambiguous eligibility sentence** you had to adjudicate.
- ▶ **Stance:** a documented N you can defend beats a matched N you cannot. P1 is **optional and ungraded** — no separate deadline and nothing to submit; keep the funnel and its **one hardest judgment call** in your group GitHub repo, and be ready to defend that call.